

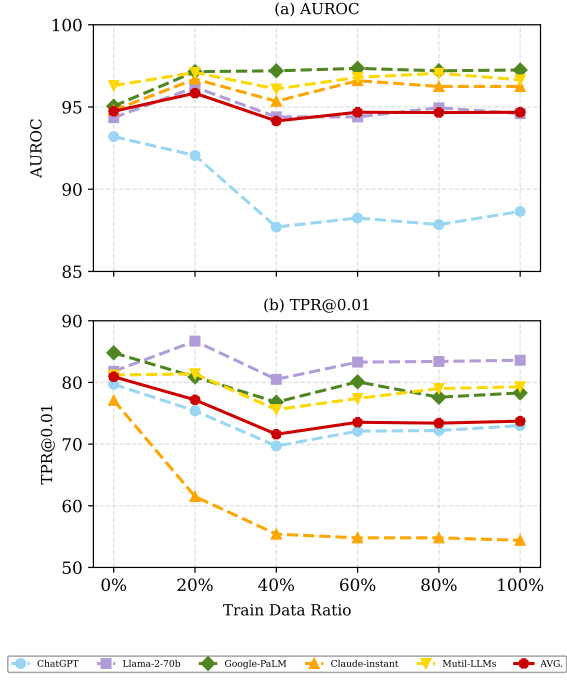


Figure 10: Performance of RepreGuard in Terms of AUROC and TPR@0.01 on 1000 “HWT-LGT” Pairs from 4 Different LLMs after LGT Pretraining. The raw text generated by each model is used to model representation features and set thresholds.

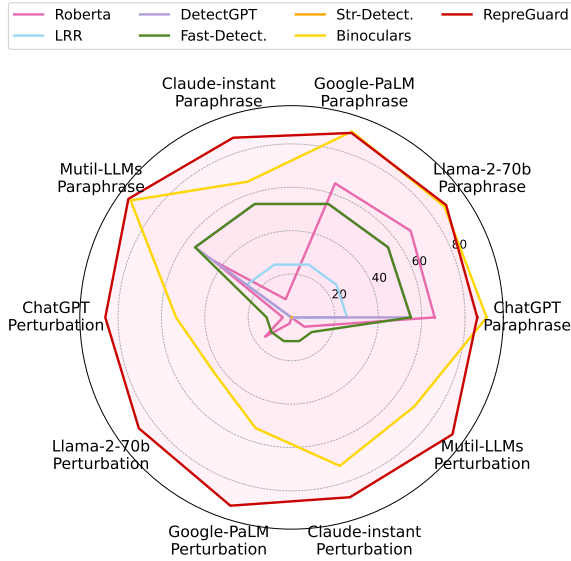


Figure 11: Performance Comparison of RepreGuard on Paraphrase and Perturbation Attack in Terms of TPR@0.01 on 1000 “HWT-LGT” Pairs from 4 Different LLMs. The raw text generated by each model is used to model representation features and set thresholds.

that the hidden representation contains rich information, offering significant advantages in hallucination detection tasks.

However, our method, RepreGuard, systematically identifies differences in hidden states between LGT and HWT to distinguish them. Similar to MGCT, RepreGuard focuses on disparities within the hidden space, while MGCT (Monea et al., 2024) places greater emphasis on causal intervention and mechanistic explanation. In contrast to Azaria and Mitchell (2023), RepreGuard does not require training additional networks, enabling efficient detection within an unsupervised framework. Furthermore, compared to LLM-Check (Sriramanan et al., 2024), which relies on the covariance features of generation, RepreGuard is designed to capture the hidden representations underlying behavioral processes, allowing the model to simulate the writing process and thereby discern differences in hidden representations between HWT and LGT.

A.5 Figure of TPR@.01 on Paraphrase & Perturbation Attack

Figure 11 illustrates the performance comparison of RepreGuard under paraphrase and perturbation attacks in terms of TPR@0.01.

Score) as a quantitative metric for hallucination detection. These methods collectively validate